PRACTICAL 10

Internet With Routers & Design Configuration of Wireless Router, DHCP Servers and Internet Cloud

PART A – Internetworking Using Router

AIM

To design and configure a simple internetwork using a router in Cisco Packet Tracer.

PROCEDURE

1. Open Cisco Packet Tracer.

2. Add 1 Router and 2 PCs.

3. Connect the PCs to the router using Copper Straight-Through Cable.

4. Open Router CLI and configure:

```
enable
config t
interface FastEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
interface FastEthernet0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
```

5. Configure PC0:

```
IP: 192.168.10.2
Subnet Mask: 255.255.255.0
Gateway: 192.168.10.1
```

6. Configure PC1:

```
IP: 192.168.20.2
Subnet Mask: 255.255.255.0
Gateway: 192.168.20.1
```

7. Send a Simple PDU from PC0 to PC1 and check the connectivity.

OBSERVATION

PC0 and PC1 were configured with different network addresses. The router successfully connected both networks and the PDU was successfully transferred.

RESULT

Thus, the internetwork using a router was successfully designed and configured, and connectivity between PC0 and PC1 was verified.

PART B – Wireless Router, DHCP Server and Internet Cloud

AIM

To design and configure an internetwork using a wireless router, DHCP server and Internet cloud.

PROCEDURE

1. Open Cisco Packet Tracer.

2. Add PC, Wireless Router, Laptop, Cable Modem, Internet Cloud and Cisco.com Server.

3. Connect the devices using the required cables.

4. Configure the wireless router.

5. Set the wireless network name (SSID) as HomeNetwork.

6. Enable DHCP on the wireless router.

7. Configure the laptop and connect it to HomeNetwork.

8. Configure the PC to obtain its IP address using DHCP.

9. Configure the Internet Cloud.

10. Configure the Cisco.com server as a DHCP and DNS server.

11. On the PC, use:

```
ipconfig /release
ipconfig /renew
```

12. Check whether the PC receives an IP address in the 192.168.0.x range.

13. Test the connection using:

```
ping Cisco.com
```

Lab manual configuration: Wireless router IP address: 192.168.0.1; Cisco.com/DNS server IP address: 208.67.220.220.

OBSERVATION

The wireless router successfully provided IP addresses to the devices using DHCP. The PC and laptop were connected to the wireless network, and connectivity to the Cisco.com server was successfully tested.

RESULT

Thus, the internetwork using a wireless router, DHCP server and Internet cloud was successfully designed and configured, and network connectivity was verified.